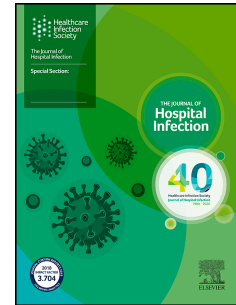


Journal Pre-proof

Comparison of Long-term and Perioperative Complications between Tunneled and Conventional Peripherally Inserted Central Catheters in Patients with Tumor: A Systematic Review and Meta-Analysis

Xiaoqing Ren, Hua Liu, Su Chen, Chengyan Han, Shuo Li, Xue Zhang



PII: S0195-6701(26)00192-1

DOI: <https://doi.org/10.1016/j.jhin.2026.05.019>

Reference: YJHIN 7865

To appear in: *Journal of Hospital Infection*

Received Date: 17 March 2026

Revised Date: 4 May 2026

Accepted Date: 7 May 2026

Please cite this article as: Ren X, Liu H, Chen S, Han C, Li S, Zhang X, Comparison of Long-term and Perioperative Complications between Tunneled and Conventional Peripherally Inserted Central Catheters in Patients with Tumor: A Systematic Review and Meta-Analysis, *Journal of Hospital Infection*, <https://doi.org/10.1016/j.jhin.2026.05.019>.

This is a PDF of an article that has undergone enhancements after acceptance, such as the addition of a cover page and metadata, and formatting for readability. This version will undergo additional copyediting, typesetting and review before it is published in its final form. As such, this version is no longer the Accepted Manuscript, but it is not yet the definitive Version of Record; we are providing this early version to give early visibility of the article. Please note that Elsevier's sharing policy for the Published Journal Article applies to this version, see: <https://www.elsevier.com/about/policies-and-standards/sharing#4-published-journal-article>. Please also note that, during the production process, errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

© 2026 The Author(s). Published by Elsevier Ltd on behalf of The Healthcare Infection Society.

Comparison of Long-term and Perioperative Complications between Tunneled and Conventional Peripherally Inserted Central Catheters in Patients with Tumor: A Systematic Review and Meta-Analysis

Xiaoqing Ren^a Hua Liu^b Su Chen^a Chengyan Han^c Shuo Li^c Xue Zhang^{a*}

^a Department of Breast Surgery, Heping Hospital Affiliated to Changzhi Medical College, Changzhi, China

^b Department of Hematology, Heping Hospital Affiliated to Changzhi Medical College, Changzhi, China

^c School of Rehabilitation, Hangzhou Medical College, Hangzhou, China

*Correspondence: Xue Zhang (zhangxue@hp1946.cn)

Summary

Background: Peripherally inserted central catheters (PICC) is widely used for long-term venous access in cancer patients, but conventional PICC is associated with high rates of infection and other complications.

Aim: To comprehensively compare the long-term postoperative complications and perioperative indicators between tunneled and conventional PICC in patients with tumor.

Methods: Studies comparing tunneled and conventional PICC from five databases were included. Meta-analyses were conducted to calculate odds ratios (OR) or standardized mean differences (SMD) with 95% confidence intervals (CI). Subgroup and sensitivity analyses were also performed.

Results: Twelve studies involving 5149 patients were included. Compared with conventional PICC, tunneled PICC significantly reduced the risks of postoperative infection (OR = 0.37, 95% CI: 0.27-0.51, $p < 0.001$), thrombosis (OR = 0.32, 95% CI: 0.22-0.47, $p < 0.001$), catheter dislodgement (OR = 0.50, 95% CI: 0.32-0.76, $p = 0.001$), unplanned extubation (OR = 0.45, 95% CI: 0.26-0.78, $p = 0.004$), and skin injury (OR = 0.69, 95% CI: 0.50-0.95, $p = 0.02$). In terms of perioperative indicators, tunneled PICC had a higher first puncture success rate (OR = 2.53, 95% CI: 1.27-5.07, $p = 0.009$) and lower 24-hour oozing risk (OR = 0.41, 95% CI: 0.31-0.53, $p < 0.001$). Subgroup analysis confirmed the stability of the infection-reducing effect.

Conclusions: Tunneled PICC offers significant advantages in reducing complications and improving perioperative performance. These findings suggest that tunneled PICC may be a valuable option for cancer patients. However, given the off-label nature of this technique and the geographic limitations of the current evidence, the results should be interpreted with caution.

Keywords: peripherally inserted central catheter; PICC; complication; oncology nursing; meta-analysis; systematic review

1. Introduction

Peripherally Inserted Central Catheters (PICC) has been widely used in oncology nursing practice for cancer patients undergoing chemotherapy, as well as for patients requiring long-term nutritional support and critical care management, leveraging their advantages of convenient operation and prolonged indwelling time [1-3]. However, conventional PICC still faces numerous complications in clinical practice, such as catheter-related infections, thrombosis, and catheter dislodgement. These complications not only may lead to treatment interruption, increased patient suffering, and higher medical costs but also can be life-threatening in severe cases [4-5]. Studies have reported that the incidence of catheter-related bloodstream infections (CRBSI) associated with conventional PICC can reach 6% in patients with hematological malignancies, while the incidence of thrombosis after PICC insertion is as high as 1%-35.5%, significantly compromising the safety and continuity of treatment [1, 6-8].

To address these unmet clinical nursing needs, tunneled PICC has emerged as an improved technique. Its core design involves creating a 2-3 cm subcutaneous tunnel before catheter insertion, separating the catheter exit site from the puncture point to form a physical barrier that prevents the invasion of skin-resident flora. Additionally, its more stable catheter fixation further alleviates the complication burden of conventional PICC [9]. In recent years, the application of tunneled PICC has expanded to cancer chemotherapy and venous therapy for critically ill patients, meeting the needs of patients' treatment and garnering particular attention in populations at high risk of infection and thrombosis [10-12]. Furthermore, the combination of precise catheterization technologies such as ultrasound guidance and intracavitary electrocardiogram (IC-ECG) positioning has enhanced the operational safety of tunneled PICC, facilitating their increasing application in cancer patients and those with poor vascular conditions [13]. Despite numerous clinical studies investigating the differences in complication rates between tunneled and conventional PICC, inconsistencies in research conclusions persist due to heterogeneities in sample size, study population, and operational protocols across studies [14-15]. Existing meta-analyses have limited their scope to a few specific complications (e.g., infections), lacking comprehensive statistical comparisons of outcomes and systematic evidence regarding perioperative indicators [9, 16].

Therefore, this study aims to comprehensively compare the differences in long-term postoperative complications and perioperative indicators between tunneled and conventional PICC in patients with tumor. The objective is to clarify the clinical advantages and application value of tunneled PICC, thereby providing high-quality evidence-based medical support for the clinical selection of PICC insertion techniques.

2. Materials and Methods

This study was conducted in accordance with the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) statement and recommendations from the Cochrane Collaboration. Additionally, it has been registered on the International Prospective Register of Systematic Reviews (PROSPERO, registration number: CRD420251249242).

2.1 Search Strategy

We searched five electronic databases, including PubMed, Embase, Web of Science, the Cochrane Library, and Scopus, from their inception to December 21, 2025. Search terms included "peripherally inserted central catheter", "PICC", "tunnel", and "tunneled". The detailed search strategy is provided in Table S1.

2.2 Selection Criteria

Inclusion criteria: (i) Study design: Cohort study or randomized controlled trial (RCT); (ii) Study population: Patients with tumors requiring PICC placement; (iii) Intervention: Tunneled PICC, defined as a PICC placed via a subcutaneous tunnel created between the venipuncture site and the catheter exit site, where the tunnel separates the two sites by a certain distance (typically ≥ 2 cm), with or without a subcutaneous cuff; (iv) Comparison: Conventional PICC; (v) Outcome indicators: Perioperative or postoperative complications associated with PICC relevant to oncology nursing outcomes.

Exclusion criteria: (i) Incomplete data availability; (ii) Incompatible statistical formats for outcome indicators; (iii) Studies not focusing on human subjects or lacking relevance to oncology nursing practice.

2.3 Outcome Indicators

Primary outcomes: Comparison of postoperative complications between conventional and tunneled PICC, including infection, thrombosis, catheter dislodgement, skin injury, phlebitis, catheter occlusion and unplanned extubation, expressed as odds ratio (OR) with 95% confidence interval (CI).

Secondary outcomes: Comparison of perioperative indicators, including intraoperative blood loss, operation time, oozing, first-attempt success rate, and pain, expressed as standardized mean difference (SMD) or OR with 95% CI.

2.4 Data Extraction

Study selection and data retrieval were performed separately by two reviewers to ensure objectivity. Disagreements were resolved through group discussion. The extracted information included first author, publication year, country of origin, study design, sample size, age, gender, disease type, catheterization technique, and outcome indicators.

2.5 Quality Evaluation

Cohort studies were assessed using the Newcastle-Ottawa Quality Assessment Scale (NOS), which includes three domains (Selection, Comparability, Outcome) with nine scoring items (1 point per item, total score 1-9). Studies with scores of 7-9 were considered high-quality, 5-6 moderate-quality, and 1-4 low-quality. RCTs were evaluated using the Cochrane Collaboration's Risk of Bias Tool 2.0 (ROB 2.0), covering five bias domains, including bias arising from the randomization process, bias due to deviations from intended interventions, bias in measurement of the outcome, bias due to missing outcome data, and bias in selection of the reported result. The final risk assessment was categorized into three levels: Low risk, High risk, and Some concerns.

2.6 Statistical Analysis

Meta-analyses were performed using RevMan 5.3 software (Cochrane Collaboration, Copenhagen, Denmark) and Stata 16.0 software (StataCorp LLC, College Station, TX, USA). Binary variables were synthesized using OR with 95% CI, and continuous variables using SMD with 95% CI. Heterogeneity was assessed using Cochran's Q test and I^2 statistic. If $p < 0.1$ and $I^2 > 50\%$, significant heterogeneity was considered, and a random-effects model was adopted; otherwise, a fixed-effects model was used. Subgroup analysis was conducted for infection risk in primary outcomes to explore sources of heterogeneity. Sensitivity analysis was performed using the one-study removal method to test the robustness of results. Publication bias was evaluated using Egger's test and funnel plots, with the trim-and-fill method applied for correction if necessary. Statistical significance was set at $p < 0.05$.

3. Results

3.1 Literature Search Results

A total of 1052 relevant studies were retrieved from the five databases. After removing 574 duplicates, 478 studies underwent title and abstract screening. Among these, 141 were non-research articles, 298 were unrelated to the study objective, 9 were conference abstracts, and 3 involved non-human subjects, leaving 27 studies for full-text review. Ultimately, 1 study with incomplete data, 12 with incompatible intervention methods, and 2 with unqualified outcome indicators were excluded. Totally, 12 studies were included in the meta-analysis, consisting of 4 cohort studies [10, 15, 17, 19] and 8 randomized controlled trials (RCTs) [11-14, 18, 20-22] (Fig. 1).

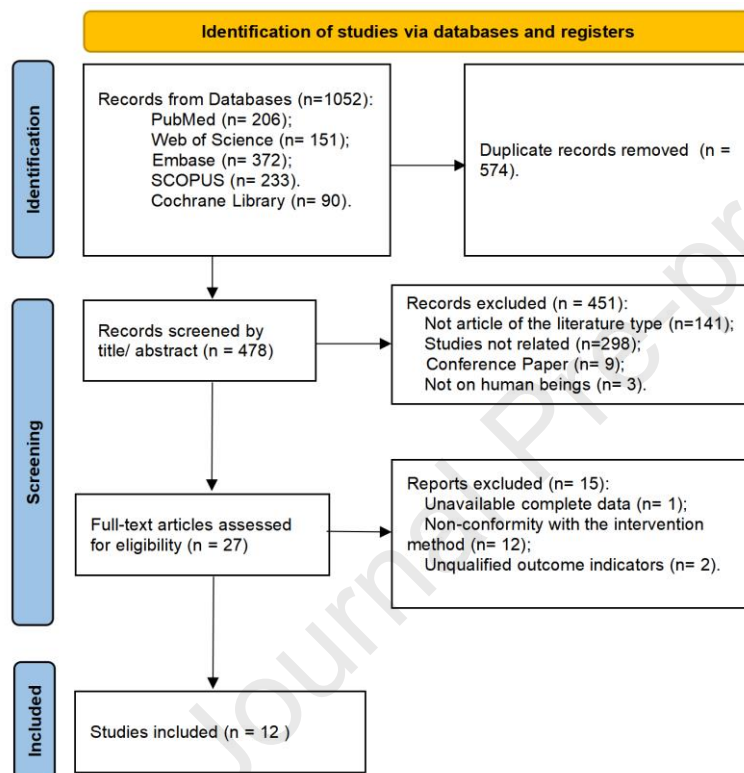


Fig. 1 Flow chart of study selection process.

3.2 Study Characteristics

A total of 5149 patients were included, with 2576 in the tunneled PICC group and 2573 in the conventional PICC group. The mean age ranged from 40 to 70 years, and males accounted for 44.1%. Among the 12 studies, 2 were from South Korea, 1 from Greece, and 9 from China. Seven studies included only tumor patients, while the remaining 5 included both tumor and non-tumor patients. Regarding PICC positioning technology, 4 studies adopted ultrasound combined with IC-ECG guidance, and 8 used ultrasound guidance alone. Eleven studies reported postoperative complications, and 10 reported perioperative complications. The study characteristics are summarized in Table 1.

Regarding the operational characteristics of tunneled PICC, substantial variability was observed across the 12 included studies. Tunnel length was reported in eight studies, ranging from 2 cm to 7 cm. Tunnel creation technique was described in ten studies, with five using sharp dissection, four using blunt dissection, and one study using both techniques. The use of a peel-away introducer

was reported in six studies. Two studies explicitly reported the absence of a subcutaneous cuff, while the remaining 10 studies did not report on cuff use. PICC specifications (including brand, model, French size, and lumen number) were reported in nine studies. A detailed summary of these technical characteristics is presented in Table 2.

3.3 Methodological Quality Assessment

All 4 cohort studies[10,15,17,19] achieved NOS scores ≥ 8 , indicating high quality (Table 3). For the 8 RCTs, ROB 2.0 assessment revealed "Some concerns" for 2 studies[12,20] and "Low risk" for 6 [11,13,14,18,21,22] (Fig. 2).

Table 1 Study characteristics.

Study	Country	Study Type	No. of T-PICC/ C-PICC	Age of T-PICC/ C-PICC	Male (%)	Disease	Localization technique of PICC	Outcomes
Lu 2025 [10]	China	Cohort	113/113	60.5±13.1/62.0±13.2	53.1	tumor	ultrasound+IC-ECG	1-1; 1-2; 1-3; 1-4; 1-6; 1-7
Kim 2025 [11]	South korea	RCT	651/650	68.0±15.9/68.7±14.6	50.3	tumor and non-tumor	ultrasound	1-1; 1-5; 2-2
Maria 2019 [12]	Greece	RCT	31/29	45.6±11.6/48.0 ±12.0	50.0	tumor and non-tumor	ultrasound	1-1; 1-5; 2-4; 2-5
Sheng 2025 [13]	China	RCT	302/303	65.0±15.0/66.0±15.0	17.2	tumor and non-tumor	ultrasound	1-1; 1-2; 1-3
Dai 2020 [14]	China	RCT	87/87	55.4±12.5/54.8±13.9	60.9	tumor	ultrasound+IC-ECG	1-1; 1-2; 1-3; 1-4; 1-6; 1-7; 2-1; 2-2; 2-5
Kim 2019 [15]	South korea	Cohort	302/309	45.7±11.3/45.7±11.5	45.2	tumor and non-tumor	ultrasound	1-1; 1-2; 1-3; 1-4; 2-2; 2-5
Zeng 2025 [17]	China	Cohort	79/77	53.5±8.5/54.0±8.0	51.9	tumor	ultrasound	1-1; 1-2; 2-4
Li 2024 [18]	China	RCT	277/276	59.4±11.2/58.9±10.6	60.9	tumor	ultrasound+IC-ECG	1-1; 1-2; 1-3; 1-4; 1-5; 1-7; 2-1; 2-3; 2-5
Zheng 2024 [19]	China	Cohort	100/100	53.3±12.6/52.0±12.0	51.0	tumor	ultrasound	1-3; 1-6; 1-7; 2-1; 2-3; 2-4; 2-5
Sheng 2024 [20]	China	RCT	338/338	49.6±6.0/50.3±5.8	16.7	tumor	ultrasound	1-1; 1-2; 1-3; 1-6; 1-7; 2-1; 2-2; 2-3; 2-4;
Xiao 2021 [21]	China	RCT	64/65	55.7±13.5/56.0±14.4	57.4	tumor	ultrasound+IC-ECG	1-1; 1-2; 1-3; 1-4; 1-5; 1-6; 1-7; 2-1; 2-2; 2-5
Li 2025 [22]	China	RCT	232/226	60.5±13.1/63.4±12.5	60.3	tumor and non-tumor	ultrasound	2-1; 2-2; 2-3; 2-4; 2-5

Notes: T-PICC, Tunneled Peripherally Inserted Central Catheter; C-PICC, Conventional Peripherally Inserted Central Catheter; RCT: Randomized Controlled Trial; IC-ECG, intracavitary electrocardiogram. Long-term complications: 1-1, infection; 1-2, thrombosis; 1-3, catheter dislodgement; 1-4, catheter occlusion; 1-5, unplanned extubation; 1-6, phlebitis; 1-7, skin injury. Perioperative complications: 2-1, blood loss; 2-2, operation time; 2-3, the score of pain; 2-4, first-attempt success rate; 2-5, amount of oozing.

177

178

Table 2 Operational characteristics of tunneled PICC across the 12 included studies

Study	Tunnel Length	Tunnel Creation Technique (Blunt vs. Sharp)	Use of Peel-Away Introducer	Subcutaneous Cuff	PICC Model (Brand/Model/Material/Fr/Lumen)
Lu 2025 [10]	2-3cm	Sharp	Yes	Not Reported	Bard Access Systems, USA/Groshong NXT ClearVue Catheter/Not Reported/4-Fr/single-lumen
Kim 2025 [11]	2.5cm	Sharp	Yes	No	Genoss, Korea/ UNIS PICC/polyurethane/5-Fr/two-lumen
Maria 2019 [12]	Not Reported	Sharp	Not Reported	Not Reported	Not Reported
Sheng 2025 [13]	Not Reported	Blunt	Yes	Not Reported	Branden Medical Devices, China/Not Reported/silicone/4-Fr/single-lumen
Dai 2020 [14]	3cm	Sharp	Not Reported	Not Reported	Bard Access Systems, USA /Not Reported/silicon/4-Fr/single-lumen
Kim 2019 [15]	2.5-3cm	Blunt and Sharp	Yes	No	Bard Access Systems, USA/Power PICC and Medcomp, USA/ Pro PICC/Not Reported/5-Fr/ two-lumen
Zeng 2025 [17]	Not Reported	Not Reported	Not Reported	Not Reported	Not Reported
Li 2024 [18]	5cm	Blunt	Yes	Not Reported	Branden Medical Devices, China/Groshong closed-ended valve PICC/silicon/4-Fr/single-lumen
Zheng 2024 [19]	Not Reported	Not Reported	Not Reported	Not Reported	Not Reported
Sheng 2024 [20]	3-7cm	Blunt	Not Reported	Not Reported	Branden Medical Devices, China/Proximal Valve PICC/silicone/4-Fr/single-lumen
Xiao 2021 [21]	5cm	Blunt	Not Reported	Not Reported	Bard Access Systems, USA /Power PICC SOLO 2/polyurethane/Not Reported/Not Reported
Li 2025 [22]	4cm	Sharp	Yes	Not Reported	Bard Access Systems, USA/Not Reported/ polyurethane/4-Fr/single-lumen

Table 3 Quality assessment of cohort studies with NOS scores

Item/Study	Lu 2025 [10]	Kim 2019 [15]	Zeng 2024 [17]	Zheng 2024 [19]
Selection	1.Representativeness of the exposed cohort	*	*	*
	2.Selection of the non exposed cohort	*	*	*
	3.Assessment of exposure	*	*	*
	4.Demonstration that outcome of interest was not present at start	*	*	*
Comparability	Control for important factor and additional factor	**	**	**
Outcome	1. Outcome assessment	*	*	*
	2.Follow up long enough for outcome to occur	*	*	*
	3.Adequacy of follow up	*	*	*
Total score	9	9	9	8

Notes: NOS, the Newcastle-Ottawa Quality Assessment Scale

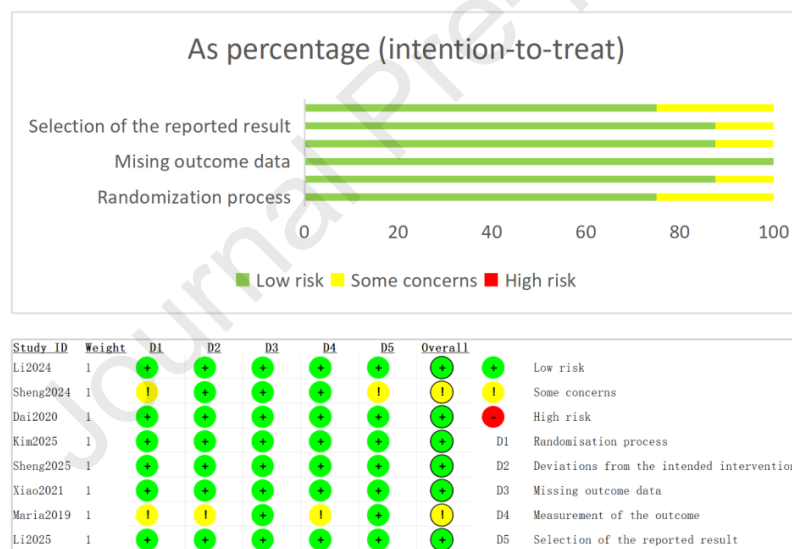


Fig. 2 Risk of bias for RCTs study by ROB2.0

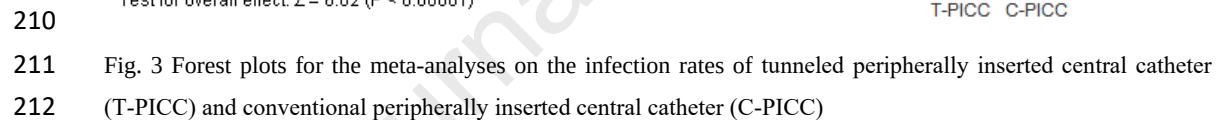
3.4 Comparison of Postoperative Complications between Conventional and Tunneled PICC

3.4.1 Infection Rate

Ten studies [10-18, 20, 21] reported postoperative infection rates, including CRBSI and local infections, such as predominantly catheter exit-site infections. Meta-analysis showed that the postoperative infection risk of tunneled PICC was significantly lower than that of conventional PICC (OR = 0.37, 95% CI: 0.27-0.51, $p < 0.001$, Fig. 3). Heterogeneity was low ($I^2 = 0\%$, $p = 0.93$), so a fixed-effects model was used.

To explore factors influencing the difference in infection risk between tunneled PICC and conventional PICC, subgroup analyses were performed by study type, disease type, PICC positioning technology, and infection type. All subgroups showed low heterogeneity ($I^2 \leq 15\%$), and tunneled PICC demonstrated consistent advantages in reducing infection risk across all subgroups (Table 4).

For the subgroup of study type, both RCTs ($P < 0.001$) and cohort studies ($P < 0.001$) confirmed a lower infection risk associated with tunneled PICC, and no significant heterogeneity was observed between subgroups ($I^2 = 11.9\%$, $P = 0.29$). Regarding disease type, tunneled PICC significantly reduced the infection risk in both tumor patients ($P < 0.001$) and mixed tumor/non-tumor patients ($P < 0.001$), with no subgroup heterogeneity detected ($I^2 = 0\%$, $P = 0.53$). For positioning technology, both ultrasound-guided ($P < 0.001$) and ultrasound combined with IC-ECG-guided ($P < 0.001$) tunneled PICC showed superior performance in lowering infection risk, and no subgroup heterogeneity was found ($I^2 = 0\%$, $P = 0.44$). For the subgroup of infection type, tunneled PICC showed a trend toward reduced infection risk in studies focusing on CRBSI ($P = 0.01$), local infection ($P = 0.13$), and combined infection ($P < 0.001$), with significant reductions observed in CRBSI and combined infection risk. No subgroup heterogeneity was noted ($I^2 = 0\%$, $P = 0.43$). In summary, the infection-reducing effect of tunneled PICC is stable and universal, unaffected by study type, disease type, positioning technology, or infection type.



214 Table 4 Subgroup analyses on the infection rates of tunneled PICC and conventional PICC

Subgroup	No. of studies	Heterogeneity		Meta-analysis		Heterogeneity between subgroup	
		<i>I</i> ²	<i>p</i> value	<i>OR</i> (95%CI)	<i>p</i> value	<i>I</i> ²	<i>p</i> value
Study type							
RCT [11-14,18,20,21]	7	0%	0.87	0.42 (0.28 - 0.63)	<0.001	11.9%	0.29
Cohort [10,15,17]	3	0%	0.98	0.29 (0.16 - 0.51)	<0.001		
Disease							
Tumor [10,14,17,18,20,21]	6	0%	0.98	0.33 (0.21 - 0.53)	<0.001	0%	0.53
Tumor and non-tumor [11-13,15]	4	0%	0.45	0.41 (0.26 - 0.64)	<0.001		
Localization technique of PICC							
Ultrasound [11-13,15,17,20]	6	15%	0.32	0.47 (0.32 - 0.70)	<0.001	0%	0.44
Ultrasound+IC-ECG [10,14,18,21]	4	0%	0.95	0.30 (0.16 - 0.56)	<0.001		
Type of Infection							

CRBSI [11,17,18]	3	0%	0.52	0.49 (0.28 - 0.84)	0.01		
Local Infection [12,14]	2	0%	0.73	0.19 (0.02 - 1.64)	0.13	0%	0.43
CRBSI+local infection [10,13,15,20,21]	5	0%	0.96	0.33 (0.22 - 0.50)	<0.001		

Notes: RCT, randomized controlled trial; PICC, peripherally inserted central catheter; IC-ECG, intracavitary electrocardiogram; CRBSI, catheter-related bloodstream infection.

3.4.2 Thrombosis, Catheter Dislodgement, Unplanned Extubation, Phlebitis, Catheter Occlusion, and Skin Injury

Postoperative thrombosis rates were documented in eight studies [10, 13-15, 17, 18, 20, 21], with the tunneled PICC group demonstrating a significantly lower thrombosis risk compared with the conventional PICC group (OR = 0.32, 95% CI: 0.22-0.47, $p < 0.001$, Fig. 4A).

Catheter dislodgement rates were investigated in another eight studies [10, 13-15, 18-21], and a marked reduction in dislodgement risk was observed in patients receiving tunneled PICC (OR = 0.50, 95% CI: 0.32-0.76, $p = 0.001$, Fig. 4B).

Four studies [11, 12, 18, 21] provided data on unplanned extubation rates, revealing that tunneled PICC were associated with a significantly decreased risk of unplanned extubation compared to conventional PICC (OR = 0.45, 95% CI: 0.26-0.78, $p = 0.004$, Fig. 4C).

Phlebitis rates were reported in five studies [10, 14, 19-21], and no statistically significant difference was detected between the tunneled and conventional PICC groups (OR = 0.32, 95% CI: 0.04-2.83, $p = 0.30$, Fig. 4D).

Data on catheter occlusion rates were available from five studies [10, 14, 15, 18, 21], and no meaningful discrepancy in occlusion risk emerged between the two groups (OR = 1.19, 95% CI: 0.71-1.97, $p = 0.51$, Fig. 4E).

Six studies [10, 14, 18-21] assessed skin injury rates, confirming that tunneled PICC conferred a significantly lower risk of skin injury relative to conventional PICC (OR = 0.69, 95% CI: 0.50-0.95, $p = 0.02$, Fig. 4F).

3.5 Comparison of Perioperative Indicators between Conventional and Tunneled PICC

Six studies reported intraoperative blood loss [14, 18-22] and operation time [11, 14, 15, 20-22], but mean and standard deviation data were unavailable for 3 studies in each outcome. Thus, meta-analysis for blood loss [18, 19, 22] and operation time [15, 20, 22] was ultimately performed with 3 studies each. No significant differences were observed between tunneled and conventional PICC in intraoperative blood loss (SMD = -0.11, 95% CI: -0.35-0.13, $p = 0.38$, Fig. 5A) or operation time (SMD = 0.22, 95% CI: -0.11-0.55, $p = 0.19$, Fig. 5B).

Four studies [18-20, 22] assessed puncture pain using the Numeric Rating Scale (NRS). No significant difference was found between the two groups (SMD = -0.05, 95% CI: -0.28-0.18, $p = 0.68$, Fig. 5C).

First puncture success rates were documented in five studies [12, 17, 19, 20, 22], with tunneled PICC exhibiting a significantly higher success rate compared to conventional PICC (OR = 2.53, 95% CI: 1.27-5.07, $p = 0.009$, Fig. 5D).

Data on oozing within 24 hours after puncture were provided by seven studies [12, 14, 15, 18, 19, 21, 22], demonstrating that tunneled PICC were associated with a significantly reduced 24-hour oozing risk (OR = 0.41, 95% CI: 0.31-0.53, $p < 0.001$, Fig. 5E).

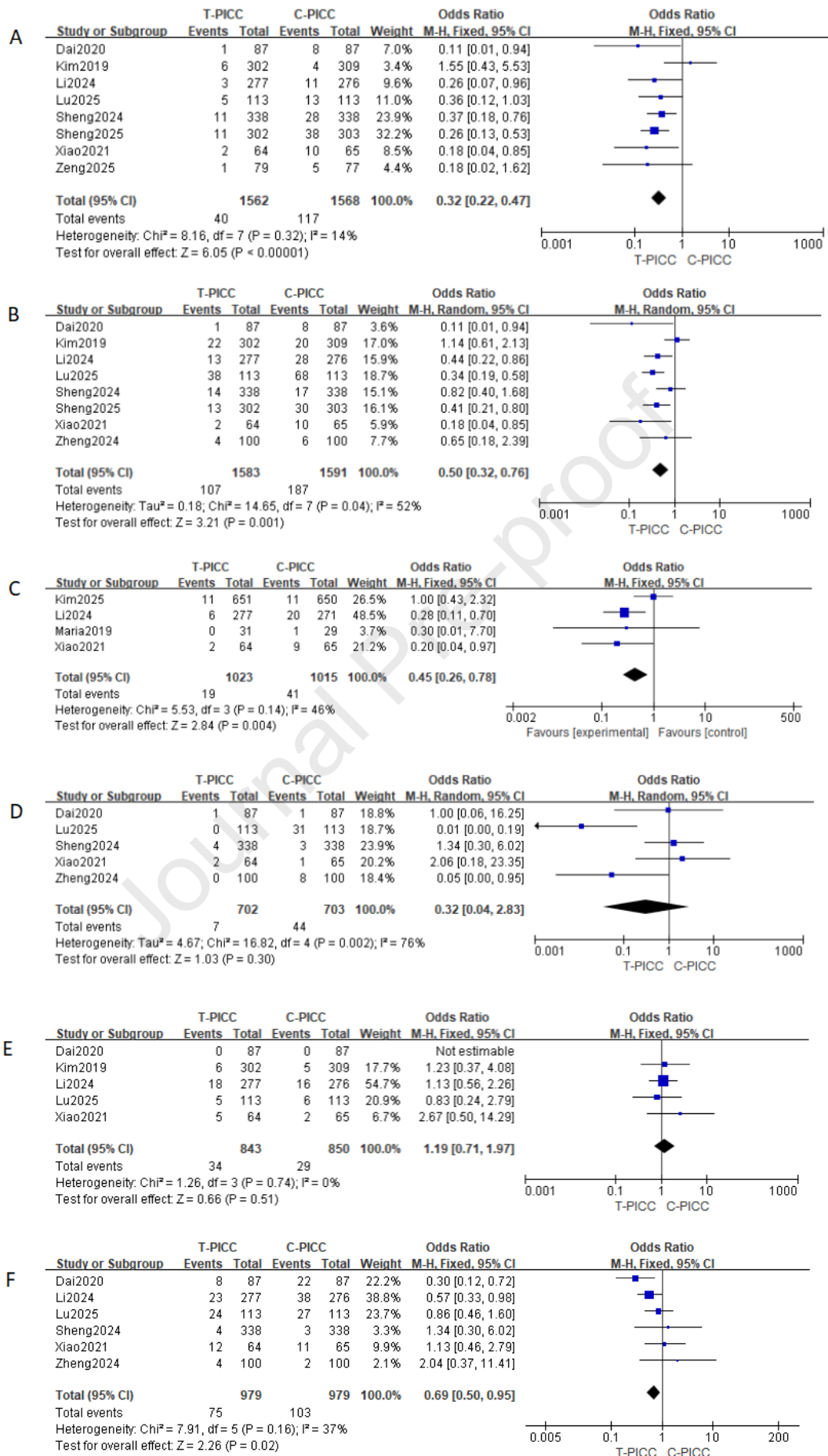


Fig. 4 Forest plots of the meta-analyses comparing postoperative complications between tunneled peripherally inserted central catheter (T-PICC) and conventional peripherally inserted central catheter (C-PICC): (A) thrombosis, (B) catheter dislodgement, (C) unplanned extubation, (D) phlebitis, (E) catheter occlusion, and (F) skin injury.

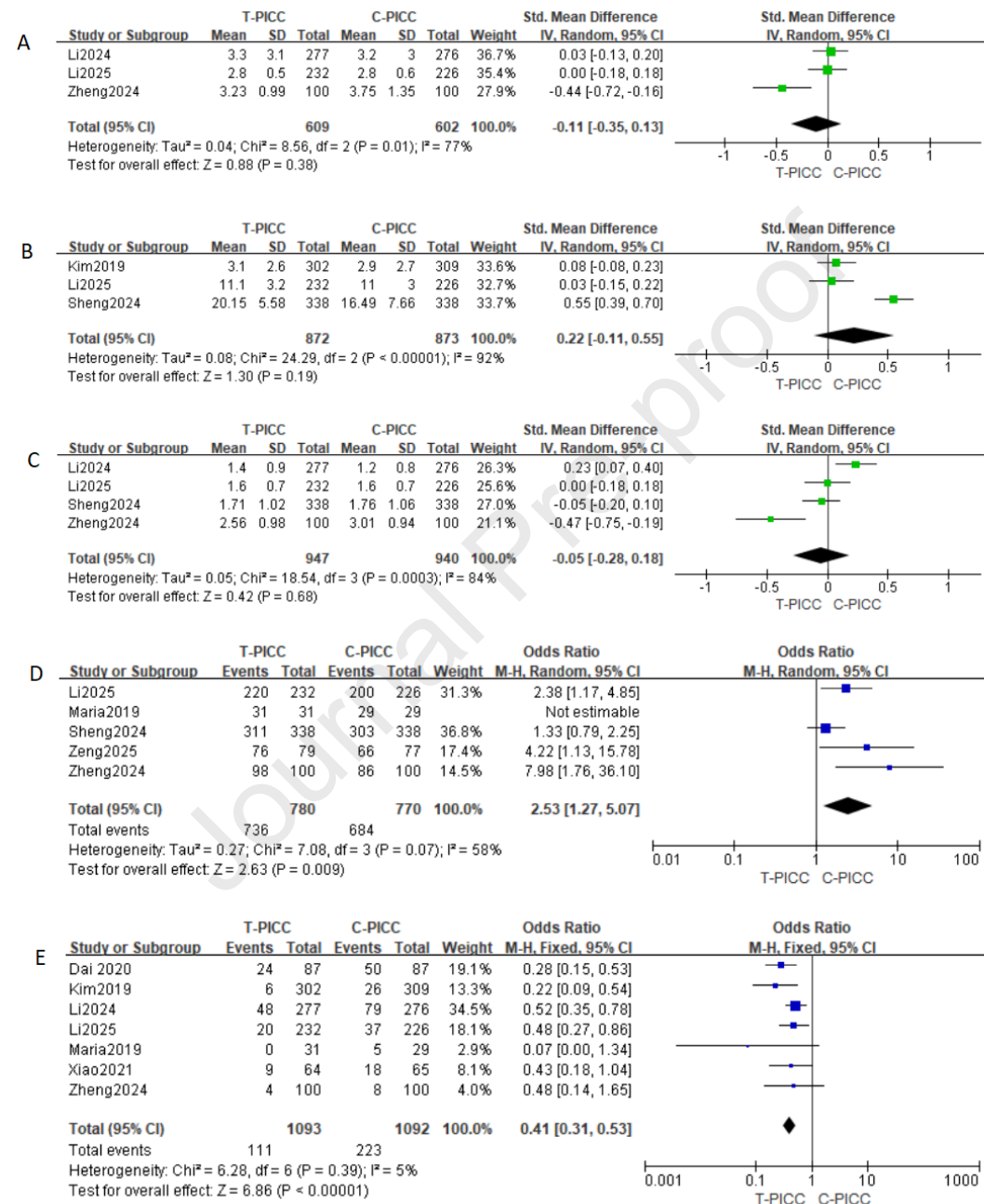


Fig. 5 Forest plots of the meta-analyses comparing perioperative indicators between tunneled peripherally inserted central catheters (T-PICC) and conventional peripherally inserted central catheters (C-PICC): (A) intraoperative blood loss, (B) operation time, (C) pain score, (D) first-attempt puncture success rate, and (E) oozing within 24 hours.

3.6 Publication Bias and Sensitivity Analysis for Primary Outcome

Publication bias for the primary outcome (postoperative infection risk) was evaluated. The funnel plot appeared asymmetrical, but Egger's test indicated no significant publication bias ($z = -0.97$, $p = 0.3311$). The trim-and-fill method was applied, and the meta-analysis conclusion remained consistent after imputing 4 hypothetical studies (OR = 0.42, 95% CI: 0.31-0.56, $p < 0.001$, Fig. 6), confirming the robustness of the results.

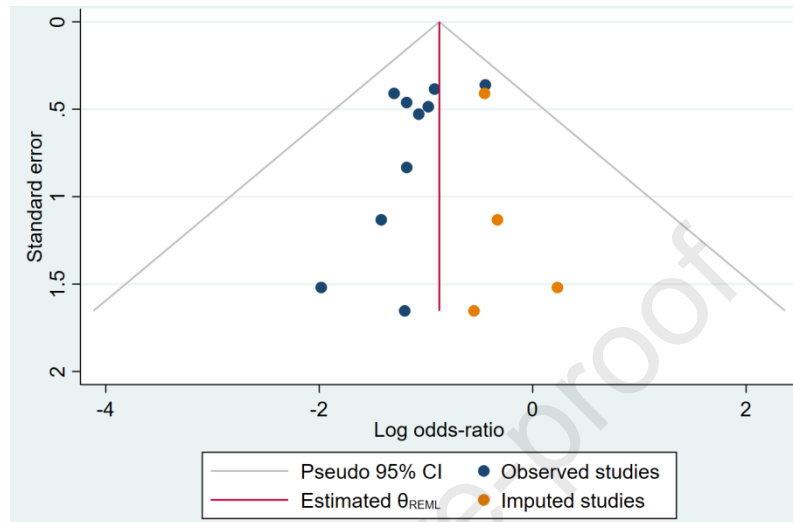


Fig. 6 Funnel plots and trim-and-fill method on infection rates of tunneled PICC and conventional PICC

Sensitivity analysis using the one-study removal method showed no significant changes in the direction or magnitude of meta-analysis results after excluding any single study, indicating stable conclusions (Table S2).

3.7 Publication Bias for Other Outcomes

Publication bias was assessed for all secondary outcomes using Egger's test (Table 5). For postoperative complications, Egger's test revealed no significant publication bias for thrombosis, catheter dislodgement, unplanned extubation, phlebitis, catheter occlusion, or skin injury (all $p > 0.05$). For perioperative indicators, Egger's test suggested potential publication bias for intraoperative blood loss and pain score (both $p < 0.05$). No significant publication bias was detected for operation time, first-attempt puncture success rate, or oozing within 24 hours (all $p > 0.05$). We performed funnel plot and trim-and-fill analysis for all outcomes (Fig. S1-S11). The funnel plot for unplanned extubation indicated that after imputing two hypothetical studies, the pooled effect became non-significant, suggesting the presence of publication bias. For phlebitis, intraoperative blood loss, and operation time, the trim-and-fill analysis could not be effectively performed due to the limited number of included studies and substantial heterogeneity. Therefore, given the constraints imposed by the small number of included studies, the results of Egger's test should be interpreted with caution.

Table 5 Egger's test for other outcomes

Outcomes	Z Value	p Value
Postoperative Complications		
Thrombosis	-0.57	0.5719
Catheter dislodgement	-1.34	0.1803

Unplanned extubation	-1.02	0.3081
Phlebitis	-1.21	0.2273
Catheter occlusion	0.29	0.7696
Skin injury	1.31	0.1901
Perioperative Indicators		
Intraoperative blood loss	-2.92	0.0035
Operation time	-0.95	0.3428
Pain score	-2.07	0.0380
First-attempt puncture success rate	1.22	0.2231
Oozing within 24 hours	-1.62	0.1056

4. Discussion

This systematic review and meta-analysis comprehensively compared the differences in postoperative complications and perioperative indicators between tunneled and conventional PICC by systematically searching five international databases. The results demonstrated that tunneled PICC provided significant advantages in reducing postoperative infection, thrombosis, catheter dislodgement, unplanned extubation, and skin injury rates. Additionally, tunneled PICC exhibited higher first puncture success rates and lower 24-hour oozing risk during the perioperative period, with no significant differences in operation time, blood loss, or puncture pain compared to conventional PICC. These findings provide reliable evidence-based support for optimizing vascular access selection in clinical oncology supportive care.

4.1 Key Findings and Clinical Significance of Postoperative Complications

Infection is one of the most severe PICC-related complications and a leading cause of catheter removal. This meta-analysis confirmed that tunneled PICC significantly reduced postoperative infection risk compared to conventional PICC, and subgroup analysis revealed this advantage was unaffected by study type, disease type, positioning technology of PICC, or infection type.

From a mechanistic standpoint, the subcutaneous tunnel acts as a physical barrier that blocks the migration of skin-resident flora along the external catheter surface, thereby effectively reducing infection rates. In the disease subgroup, cancer patients are at high risk of catheter-related infections due to chemotherapy-induced immunosuppression and mucosal barrier damage [23]. Tunneled PICC addresses this vulnerability by separating the catheter exit site from the puncture point, reducing contact between oozing at the puncture site and bacteria, and lowering mucosal injury risk from catheter dislodgement. For non-cancer patients, like critically ill patients, infection risk primarily stems from impaired skin integrity and poor local blood circulation due to prolonged bed rest. The stable fixation of tunneled PICC reduces catheter-related frictional skin injury, while the closed environment of the subcutaneous tunnel inhibits bacterial colonization [24]. Regarding positioning technology, ultrasound guidance improves puncture accuracy and reduces vascular injury, while IC-ECG guidance further optimizes the precision of catheter tip positioning, avoiding mechanical irritation and thrombosis caused by malposition near the right atrium or venous valves [25]. In the infection type subgroup, CRBSI is predominantly caused by Gram-positive bacteria from the skin surface that invade the bloodstream via biofilm formation on the catheter outer wall [26]. The subcutaneous tunnel

disrupts the continuous path for biofilm formation and reduces catheter-skin contact area, lowering bacterial adhesion. Local infections are directly associated with oozing at the puncture site and bacterial contamination from loose dressings. Tunneled PICC, with its exit site distant from the puncture point and more stable fixation, reduce dressing displacement and bacterial contamination, thereby preventing both infection types [14]. The lack of statistical significance in the local infection subgroup may be attributed to the small number of included studies and insufficient sample size, though the trend aligns with mechanistic expectations. Furthermore, the extremely low heterogeneity in infection-related analyses, combined with robust sensitivity and publication bias tests, further confirms the clear infection-preventive advantage of tunneled PICC.

Regarding infection definition heterogeneity, we acknowledge that the included studies adopted different diagnostic standards. For CRBSI, five studies followed Centers for Disease Control and Prevention/National Healthcare Safety Network (CDC/NHSN) criteria [11,13,15,17,21], one used Infusion Nurses Society (INS) criteria [18], and one used Infectious Diseases Society of America (IDSA) criteria [20]. For local infections, only three studies [10,15,21] explicitly defined this outcome using CDC criteria, while the remaining four studies [12-14,20] adopted varying definition standards. Subgroup analyses based on definition standards were not feasible due to insufficient sample sizes, which would lack statistical power and risk misleading results. Therefore, we used infection type as the primary approach to address heterogeneity, which directly responds to the concern of pooling distinct infection entities. Future studies should adopt standardized definitions for infectious outcomes to ensure comparability and homogeneity across studies.

Operational techniques of tunneled PICC placement may also contribute to outcome heterogeneity. For instance, tunnel length is considered an influencing factor. Some researchers suggest that longer tunnels may provide better bacterial barrier and catheter stability, whereas other studies have shown that tunnel length does not affect infection rates [27]. Regarding tunnel creation technique, sharp dissection may increase tissue trauma and bleeding risk, while blunt dissection helps reduce blood oozing. Furthermore, the use of a peel-away introducer may affect insertion smoothness and the risk of vessel injury. Cuffed catheters can reduce infection risk by promoting tissue ingrowth and forming a physical barrier [28]. Additionally, different PICC model specifications are associated with infection risk, for example, multi-lumen and larger-caliber catheters increase infection risk [29,30]. Unfortunately, due to incomplete reporting of these technical parameters in the original studies, we were unable to perform prespecified subgroup analyses.

Thrombosis is another common PICC-related complication, associated with mechanical irritation of vascular endothelium by the catheter, hemodynamic changes, and abnormal catheter tip positioning [31]. This study found that tunneled PICC had a lower thrombosis rate than conventional PICC, likely due to its more stable catheterization path and higher accuracy of tip positioning, which can reduce intravascular catheter displacement and floating, minimizing sustained irritation to vascular endothelium. Other studies have shown that tunneled PICC, with its larger exit-site vein diameter, reduces thrombosis caused by repeated mechanical friction and reduced blood flow [32].

Catheter dislodgement and unplanned extubation are key factors affecting PICC indwelling time. Conventional PICC, lacking the fixation of a subcutaneous tunnel, is prone to dislodgement due to patient activity or postural changes, leading to unplanned extubation [33]. This study

showed that tunneled PICC had lower dislodgement and unplanned extubation rates, indicating that the subcutaneous tunnel design effectively enhanced catheter stability, reduced dislodgement risk, prolonged indwelling time, and lowered medical costs and patient suffering from repeated catheterization. Additionally, tunneled PICC had lower skin injury rate, possibly due to more uniform tension distribution at the catheter exit site and reduced skin irritation from fixed dressings [34].

Notably, no significant differences were observed in phlebitis or catheter occlusion rates between the two catheterization methods. Phlebitis is primarily associated with catheter material, vascular injury during insertion, and chemotherapy drug irritation [35]. Since both tunneled and conventional PICC use similar materials and require peripheral venous puncture, it is unsurprising that no advantage in phlebitis reduction was observed. However, the phlebitis meta-analysis exhibited extremely high heterogeneity and a wide confidence interval, making the pooled result difficult to interpret. This heterogeneity likely arose from inconsistent directions of effect across the included studies. The low overall incidence of phlebitis, differences in assessment criteria and follow-up duration, and variations in patient populations (e.g., tumor type, chemotherapy regimen) may have further exacerbated the heterogeneity. As for catheter occlusion, it is mainly related to blood reflux, drug deposition, and improper catheter maintenance [36]. This result suggests that standardized catheter maintenance remains critical for preventing occlusion, regardless of the catheterization method.

4.2 Comparison of Perioperative Indicators and Clinical Practice Implications

Perioperative indicators directly reflect the feasibility and patient tolerance of catheterization techniques in perioperative oncology nursing practice for PICC placement. This study found that tunneled PICC had a higher first puncture success rate and lower 24-hour oozing risk. This may be attributed to less vascular injury and more secure catheter fixation during tunneled PICC insertion, reducing patient suffering and medical resource waste from puncture failure, as well as infection risk associated with oozing [15]. It should be noted that the meta-analysis of first puncture success rate showed a wide confidence interval and substantial heterogeneity, which should be interpreted with caution. A leave-one-out sensitivity analysis revealed that exclusion of the study by Zheng et al. [19] markedly narrowed the confidence interval of the pooled estimate (after exclusion: OR = 1.82, 95% CI: 1.23-2.71, $I^2 = 43\%$), indicating that this study had a considerable impact on the overall stability and precision of the result. Several factors may explain this finding. First, the study was retrospective in design, which may introduce selection bias. Second, the conventional PICC group had a relatively low success rate (86.0%) compared with other included studies (typically 88%-92%). Third, the sample size was relatively modest compared with other included studies, which may compromise result stability. Furthermore, the study by Maria et al. [12] could not be included in this meta-analysis because both groups achieved 100% success, rendering the odds ratio incalculable and leading statistical software (RevMan) to automatically treat it as "not estimable" and exclude it from the pooled analysis. Nevertheless, based on data from only three or four studies, no significant differences were observed in operation time, blood loss, or puncture pain. Given the limited sample size and potential for insufficient power, these results warrant cautious interpretation and do not definitively confirm that tunneled PICC does not compromise patient tolerance.

4.3 Strengths and Limitations

Our study has several strengths. Firstly, this is the first meta-analysis to comprehensively

compare long-term postoperative complications and perioperative indicators between tunneled and conventional PICC, providing comprehensive evidence-based support for clinical practice in oncology nursing. Secondly, subgroup analysis, sensitivity analysis, and publication bias testing were performed for the infection rate in primary outcomes, confirming robust and reliable results. Moreover, the included studies, consisting of RCTs and cohort studies, underwent rigorous quality assessment using the NOS and ROB 2.0 tools, ensuring high overall study quality and enhancing result credibility.

However, there are some limitations in this study. First, geographical bias may exist, as most studies were conducted in the Asia-Pacific region. This geographic skew may limit the generalizability of these findings to other regions with different medical resources and clinical practices. Specifically, there are significant differences in vascular access practices between Asian and Western healthcare settings. In China, PICC insertion is predominantly performed by specialized trained nurses, whereas in other Western countries, the procedure may be performed by physicians, interventional radiologists, or dedicated vascular access teams [37,38]. Other differences include device availability, standardization of tunneling techniques, and infection surveillance protocols [39-41]. Regarding infection surveillance, there are notable differences in the application of CRBSI definitions between China and Western countries. Even when following the standard CDC/NHSN CLABSI definition, significant heterogeneity exists across different countries [42,43]. Furthermore, variations in blood culture practices, paired blood cultures, and catheter tip culture methods may directly impact reported infection rates and the estimated protective effect of the tunneling technique [44,45]. These contextual factors may influence baseline complication rates across studies and the magnitude of our pooled estimates. Therefore, our findings should be interpreted as most directly applicable to Asian healthcare systems, and validation in Western populations is needed. Besides, publication bias, although assessed, may still be present. While Egger's test and trim-and-fill analysis did not indicate major bias for most outcomes, the analysis was limited for some outcomes due to the small number of included studies, making the detection of bias less reliable. As a result, the estimated effects for outcomes with few studies, such as phlebitis, operation time and intraoperative blood loss, should be interpreted with caution. Second, incomplete data reporting precluded several planned analyses. Data on indwelling time and maintenance costs were insufficient for pooled analyses. Only two studies [14,21] reported quantifiable cost data, both from single-center Chinese studies. Both studies found that tunneled PICC had slightly higher insertion costs but lower daily maintenance costs. However, these may not be generalizable to other healthcare systems with different pricing structures and resource availability. Notably, no included study performed a formal cost-effectiveness analysis, which would be necessary to determine whether the clinical benefits of tunneled PICC justify the additional upfront costs, particularly in resource-constrained healthcare settings. Therefore, these findings should be considered preliminary, and future large-scale studies are needed to confirm the equivalence of tunneled and conventional PICC with respect to these indicators.

Despite the positive findings of this meta-analysis, it must be acknowledged that the application of subcutaneous tunneling for PICC insertion currently constitutes an off-label use of commercially available PICC devices. The vast majority of PICC products on the market have obtained Conformité Européenne (CE) marking and Food and Drug Administration (FDA) clearance for conventional percutaneous insertion following their specific Instructions for Use

(IFU), which do not include modification of the insertion technique via a subcutaneous tunnel. Lingegowda et al. described the use of a non-cuffed tunnelled PICC placed via the internal jugular vein in oncology patients as "off-label use", as the device was repurposed for an alternative insertion site and technique [46]. Notably, five of the twelve studies included in this meta-analysis enrolled mixed populations of oncology and non-oncology patients, implying that the application of tunneled PICC in critically ill patients or in the intensive care unit setting also constitutes off-label use. Unfortunately, none of the original studies analyzed the differences in benefit-risk profiles for this specific population. Therefore, clinicians and nurses considering the adoption of tunneled PICC should verify local institutional policies regarding off-label device use, obtain explicit informed consent from patients, ensure that operators have received adequate training in the modified technique, and document the rationale for this approach in the medical record.

5. Conclusion

Compared to conventional PICC, tunneled PICC offers advantages in reducing postoperative complications including infection, thrombosis, catheter dislodgement, unplanned extubation, and skin injury, while also improving first-puncture success rates and reducing perioperative oozing, without increasing operation time or patient pain. These findings suggest that tunneled PICC may represent a valuable option for patients requiring venous access, particularly those at high risk of infection, thrombosis, or catheter dislodgement. However, given the off-label nature of this technique and the geographic predominance of available evidence from Asian healthcare settings, clinicians should carefully weigh the potential benefits against institutional policies and regulatory requirements. Further validation in western populations and engagement with manufacturers for formal device approval are warranted.

CRediT authorship contribution statement

Xiaoqing Ren: Conceptualization, methodology, resources, software, writing-original draft. **Hua Liu:** Data curation, formal analysis, visualization. **Su Chen:** Methodology, validation, investigation, writing-review & editing. **Chengyan Han:** Resources, methodology, software. **Shuo Li:** Formal analysis, software, visualization, writing-original draft. **Xue Zhang:** Writing-review & editing, supervision. All authors have read and approved the final manuscript.

Ethics statement

Not required.

Data availability statement

The data supporting the findings of this study are available within the article/supplementary material.

Declaration of generative AI and AI-assisted technologies in the writing process

No AI tools/services were used during the preparation of this work.

Funding

This study received no external funding.

Declaration of competing interest

The authors declare no conflict of interest.

References

1. Guman NAM, van Dooremaal BM, Beenen LF, Scheerder MJ, Becking AM, Lagerweij MF, et al. Diagnostic accuracy of computed tomography venography compared with ultrasonography for detection of peripherally inserted central venous catheter-related thrombosis in patients with cancer: the DETECT study. *J Thromb Haemost.* 2026; 24(1):180-189. doi: 10.1016/j.jtha.2025.08.008.
2. Lee SM, Park J. Risk factors for complications associated with peripherally inserted central venous catheters for parenteral nutrition: Machine learning and survival analysis based on deep learning. *Clin Nutr.* 2025; 55:249-256. doi: 10.1016/j.clnu.2025.11.008.
3. Long G, Wang Z, Peng H, Li Y. Comparison of modified midlines versus PICCs in a medical ICU setting: A retrospective study. *Medicine (Baltimore).* 2025; 104(46):e46039. doi: 10.1097/MD.00000000000046039.
4. Liu R, Xu H, Pu L, Xie X, Chen H, Wu Z, et al. Clinical characteristics of peripherally inserted central catheter-related complications in cancer patients undergoing chemotherapy: a prospective and observational study. *BMC Cancer.* 2023; 23(1):894. doi: 10.1186/s12885-023-11413-0.
5. Dorea E, Leeson R, Zimpelman L, Humphrey J. Nursing Practices Related to Peripherally Inserted Central Catheters in the Neonatal Intensive Care Unit: Scoping Review. *J Infus Nurs.* 2025; 48(6):395-402. doi: 10.1097/NAN.0000000000000618.
6. Guo Q, Liu Y, Zhu X, Gao Y, Zhou Y, Zhang X. Development and validation of a nomogram for predicting PICC catheter-related bloodstream infection among patients with hematologic malignancies. *BMC Infect Dis.* 2025; 25(1):1370. doi: 10.1186/s12879-025-11815-4.
7. Rabelo-Silva ER, Saffi MAL, Hirakata VN, Ferro EB, Lourenço SA, da Silva AIM, et al. External Validation of the Michigan Risk Score for Predicting Peripherally Inserted Central Catheter-Related Deep Vein Thrombosis: A Multicenter Study in Brazil. *J Infus Nurs.* 2026; 49(1):43-51. doi: 10.1097/NAN.0000000000000623.
8. Petitgas P, Bun R, Tounkara B, Simac C, Bertolotti A, Poubeau P, et al. Risk factors for deep vein thrombosis during peripherally inserted central catheter-related infections: A retrospective study. *Infect Dis Now.* 2025; 55(8):105181. doi: 10.1016/j.idnow.2025.105181.
9. Shim DJ, Kim ET, Lee JH, Kwon Y, Cho SB, Yoon CJ, et al. Tunneled Peripherally Inserted Central Catheters and Bloodstream Infection: A Systematic Review and Meta-Analysis. *Cardiovasc Intervent Radiol.* 2026; 49(1):9-16. doi: 10.1007/s00270-025-04289-z.
10. Lu T, Xu Y, Wei J, Ying Y. Comparison of Tunneled vs Traditional PICCs in Cancer Patients: A Propensity Score-Matched Study. *J Infus Nurs.* 2025; 48(6):422-432. doi: 10.1097/NAN.0000000000000616.
11. Kim ET, Lee JH, Shim DJ, Kwon Y, Cho SB, Kim KJ, et al. Subcutaneous tunnelling versus conventional insertion of peripherally inserted central catheters in hospitalized patients (TUNNEL-PICC): a multi-centre, open-label, randomized, controlled trial. *J Hosp Infect.* 2025; 155:106-114. doi: 10.1016/j.jhin.2024.10.008.

12. Maria K, Theodoros K, Maria B, Panagiotis K, Emmanouil S, Evangelos KA. Implementation of tunneled versus not tunneled peripherally inserted central catheters. *J Vasc Nurs.* 2019; 37(2):132-134. doi: 10.1016/j.jvn.2018.11.007.
13. Sheng Y, Gao W, Dongye SY. Effectiveness of subcutaneous tunneling technique in reducing PICC dislodgement and malposition: a pilot multicenter randomized controlled trial. *Front Nurs.* 2025; 12(1): 145-153. doi: 10.2478/FON-2025-0016
14. Dai C, Li J, Li QM, Guo X, Fan YY, Qin HY. Effect of tunneled and nontunneled peripherally inserted central catheter placement: A randomized controlled trial. *J Vasc Access.* 2020; 21(4):511-519. doi: 10.1177/1129729819888120.
15. Kim IJ, Shim DJ, Lee JH, Kim ET, Byeon JH, Lee HJ, et al. Impact of subcutaneous tunnels on peripherally inserted catheter placement: a multicenter retrospective study. *Eur Radiol.* 2019; 29(5):2716-2723. doi: 10.1007/s00330-018-5917-x.
16. Hong J, Mao X. Complications of tunneled and non-tunneled peripherally inserted central catheter placement in chemotherapy-treated cancer patients: a meta-analysis. *Front Surg.* 2024; 11:1469847. doi: 10.3389/fsurg.2024.1469847.
17. Zeng J, Zhang Y, Xu H. Comparison of the safety and efficacy of EDUG-guided tunnelled PICC versus traditional PICC in patients with cancer: A retrospective study. *Medicine (Baltimore).* 2025; 104(41):e44727. doi: 10.1097/MD.00000000000044727.
18. Li J, Hu Z, Luo M, Wu Z, Dou X, Wang Z, et al. Safety and effectiveness of tunneled peripherally inserted central catheters versus conventional PICC in adult cancer patients. *Eur Radiol.* 2024; 34(12):7776-7785. doi: 10.1007/s00330-024-10852-y.
19. Zheng Y, Xiang X, Li L, Zhang L, He S. Comparison on clinical efficacy and adverse reactions between TPICC and ultrasound-guided PICC for advanced tumors: A retrospective study. *Medicine (Baltimore).* 2024; 103(42):e38130. doi: 10.1097/MD.00000000000038130.
20. Sheng Y, Yang LH, Wu Y, Gao W, Dongye SY. Implementation of Tunneled Peripherally Inserted Central Catheters Placement in Cancer Patients: A Randomized Multicenter Study. *Clin Nurs Res.* 2023; 33(1):19-26. doi: 10.1177/10547738231194099.
21. Xiao MF, Xiao CQ, Li J, Dai C, Fan YY, Cao HJ, et al. Subcutaneous tunneling technique to improve outcomes for patients undergoing chemotherapy with peripherally inserted central catheters: a randomized controlled trial. *J Int Med Res.* 2021; 49(4):3000605211004517. doi: 10.1177/03000605211004517.
22. Li L, Wu X, Lin L, Cai Z, Ye X, Lin Y, et al. Effects of three peripherally inserted central catheters insertion techniques on catheterization outcomes: A randomized controlled trial. *Int J Nurs Stud.* 2025; 172:105209. doi: 10.1016/j.ijnurstu.2025.105209.
23. Rolston KV. Infections in Cancer Patients with Solid Tumors: A Review. *Infect Dis Ther.* 2017; 6(1):69-83. doi: 10.1007/s40121-017-0146-1.
24. Li J, Wu X, Li Z, Zhou X, Cao J, Jia Z, et al. Nursing resources and major immobility complications among bedridden patients: A multicenter descriptive study in China. *J Nurs Manag.* 2019; 27(5):930-938. doi: 10.1111/jonm.12731.
25. Zhang L, Wang M, Zhao M, Pu S, Zhao J, Zhu G, et al. Efficacy and safety of intracavitary electrocardiography-guided peripherally inserted central catheters in pediatric patients: a systematic review and meta-analysis. *PeerJ.* 2024; 12:e18274. doi: 10.7717/peerj.18274.
26. Li S, Xie Y, Liu W, Wan Y, Zuo J, Yi H. A single-center study on epidemiology, microbial distribution, and mortality predictors in catheter-related bloodstream infections. *BMC Infect Dis.* 2025; 26(1):17. doi: 10.1186/s12879-025-12249-8.

27. Li S, Lu H, Li S, Peng Y, Guo H, Yi T, et al. Impact of subcutaneous tunnel length on infection risk in tunneled PICCs: a study in cancer patient. *Sci Rep.* 2025; 15(1):5430. doi: 10.1038/s41598-024-82430-7.
28. Sze Yong T, Vijayanathan AA, Chung E, Ng WL, Yaakup NA, Sulaiman N. Comparing catheter related bloodstream infection rate between cuffed tunnelled and non-cuffed tunnelled peripherally inserted central catheter. *J Vasc Access.* 2022; 23(2):225-231. doi: 10.1177/1129729820987373.
29. Trautman K, Kobayashi T, McPherson C, Brust K. Defaulting peripherally inserted central catheter (PICC) orders to single lumen: A quality improvement initiative at a Midwestern academic center 2022-2023. *Antimicrob Steward Healthc Epidemiol.* 2023; 3(1):e242. doi: 10.1017/ash.2023.511.
30. Faltoni M, Catho G, Pianca E, Minka-Obama B, Zanella MC, Chraiti MN, et al. Catheter size and risk of short-term peripheral venous catheter-associated bloodstream infections: an observational study. *Clin Microbiol Infect.* 2024; 30(4):548-551. doi: 10.1016/j.cmi.2023.12.018.
31. Wang XX, He Y, Chu J, Xu JS. Risk factors analysis and the establishment of nomogram prediction model for PICC-related venous thrombosis in patients with lymphoma: a double-center cohort-based case-control study. *Front Oncol.* 2024; 14:1347297. doi: 10.3389/fonc.2024.1347297.
32. Kagan E, Salgado CD, Banks AL, Marculescu CE, Cantey JR. Peripherally inserted central catheter-associated bloodstream infection: Risk factors and the role of antibiotic-impregnated catheters for prevention. *Am J Infect Control.* 2019; 47(2):191-195. doi: 10.1016/j.ajic.2018.07.006.
33. Zhang J, Ma G, Peng S, Hou J, Xu R, Luo L, et al. Risk Factors and Predictive Models for Peripherally Inserted Central Catheter Unplanned Extubation in Patients With Cancer: Prospective, Machine Learning Study. *J Med Internet Res.* 2023; 25:e49016. doi: 10.2196/49016.
34. Li J, Qian Q, Nan J, Li W, Zhang T, Zhang H, et al. The prevalence and risk factors of medical adhesive-related skin injury in cancer patients with peripherally inserted central catheter: A systematic review and meta-analysis. *J Vasc Access.* 2025; 11297298251319824. doi: 10.1177/11297298251319824.
35. Cernuda Martínez JA, Suárez Mier MB, Martínez Ortega MDC, Casas Rodríguez R, Villafranca Renes C, Del Río Pisabarro C. Risk factors and incidence of peripheral venous catheters-related phlebitis between 2017 and 2021: A multicentre study (Flebitis Zero Project). *J Vasc Access.* 2024; 25(6):1835-1841. doi: 10.1177/11297298231189963.
36. Mitbander UB, Geer MJ, Taxbro K, Horowitz JK, Zhang Q, O'Malley ME, et al. Patterns of use and outcomes of peripherally inserted central catheters in hospitalized patients with solid tumors: A multicenter study. *Cancer.* 2024; 128(20):3681-3690. doi: 10.1002/cncr.34410.
37. Choi K, Keum MA, Kim MS, Kim Y, Choi S, Kyoung KH, et al. Feasibility of the Ultrasound-Guided Insertion of the Peripherally Inserted Central Catheter (PICC) by the Vascular Surgeon at the Bedside in the Trauma Intensive Care Unit. *Ann Vasc Surg.* 2022; 80:143-151. doi: 10.1016/j.avsg.2021.08.038.
38. Fong NI, Holtzman SR, Bettmann MA, Bettis SJ. Peripherally inserted central catheters: outcome as a function of the operator. *J Vasc Interv Radiol.* 2001;12(6):723-9. doi: 10.1016/s1051-0443(07)61444-4.
39. Muttalib F, González-Dambrasuskas S, Lee JH, Steere M, Agulnik A, Murthy S, et al. Pediatric Emergency and Critical Care Resources and Infrastructure in Resource-Limited Settings: A Multicountry Survey. *Crit Care Med.* 2021; 49(4):671-681. doi: 10.1097/CCM.0000000000004769.
40. Kalma B, van Zundert A. Perils of the PICC: Peripherally Inserted Central Catheter-Associated Complications and Recommendations for Prevention in Clinical Practice-A Narrative Review. *Healthcare (Basel).* 2025; 13(16):1993. doi: 10.3390/healthcare13161993.
41. Nelson SE, Tsetsou S, Liang J. Leaving no culture undrawn: Time to revisit the CLABSI and CAUTI metrics. *J Crit Care.* 2024; 79:154442. doi: 10.1016/j.jcrc.2023.154442.
42. de Mooij CEM, van der Velden WJFM, Verweij PE, de Haan AFJ, van Groningen LFJ, Meijer C, et al.

- 637 Surveillance of catheter-related bloodstream infections in haemato-oncology patients: comparison of two
638 definitions. *J Hosp Infect.* 2020;105(4):686-690. doi: 10.1016/j.jhin.2020.04.027.
- 639 43. Gaur AH, Miller MR, Gao C, Rosenberg C, Morrell GC, Coffin SE, et al. Evaluating application of the
640 National Healthcare Safety Network central line-associated bloodstream infection surveillance definition: a
641 survey of pediatric intensive care and hematology/oncology units. *Infect Control Hosp Epidemiol.* 2013;
642 34(7):663-70. doi: 10.1086/671005.
- 643 44. Ombelet S, Barbé B, Affolabi D, Ronat JB, Lompo P, Lunguya O, et al. Best Practices of Blood Cultures in
644 Low- and Middle-Income Countries. *Front Med (Lausanne).* 2019; 6:131. doi: 10.3389/fmed.2019.00131.
- 645 45. Beekmann SE, Diekema DJ, Huskins WC, Herwaldt L, Boyce JM, Sherertz RJ, et al. Diagnosing and
646 reporting of central line-associated bloodstream infections. *Infect Control Hosp Epidemiol.* 2012;
647 33(9):875-82. doi: 10.1086/667379.
- 648 46. Lingegowda D, Gehani A, Sen S, Mukhopadhyay S, Ghosh P. Centrally inserted tunnelled peripherally
649 inserted central catheter: Off-label use for venous access in oncology patients. *J Vasc Access.* 2020;
650 21(5):773-777. doi: 10.1177/1129729820909028.

- The results of this meta-analysis confirm that tunneled PICCs outperform conventional PICCs in reducing key postoperative complications (infection, thrombosis, catheter dislodgement, etc.) and optimizing perioperative indicators (higher first-attempt puncture success rate, lower 24-hour oozing risk) for medium-to-long-term venous therapy.
- Subgroup analyses demonstrate that the infection-reducing advantage of tunneled PICCs is stable and universal, unaffected by study type, disease type (tumor/non-tumor), or PICC positioning technology (ultrasound alone/ultrasound + IC-ECG).
- Additionally, this study provides comprehensive evidence for clinical vascular access selection, filling the gap of previous meta-analyses that focused only on specific populations or limited complications.